whether or not there might be something outside it. Something that may ultimately destroy it all.

THE UNBEARABLE WEAKNESS OF GRAVITY

We've known for a long time that something has to be wrong with gravity. It works too well. Einstein's general relativity has so far performed perfectly in every situation in which it's been tested. For decades, physicists have tried to find some kind of deviation, somewhere, anywhere, that would show us how the simple[fn1] equations written down in Einstein's theory inevitably break down. Somewhere, in some extreme regime, like at the edge of a black hole or among the particles at the center of a neutron star, the equations must have some kind of a crack. We haven't found it in any of our searches so far, but we're sure it has to be there.

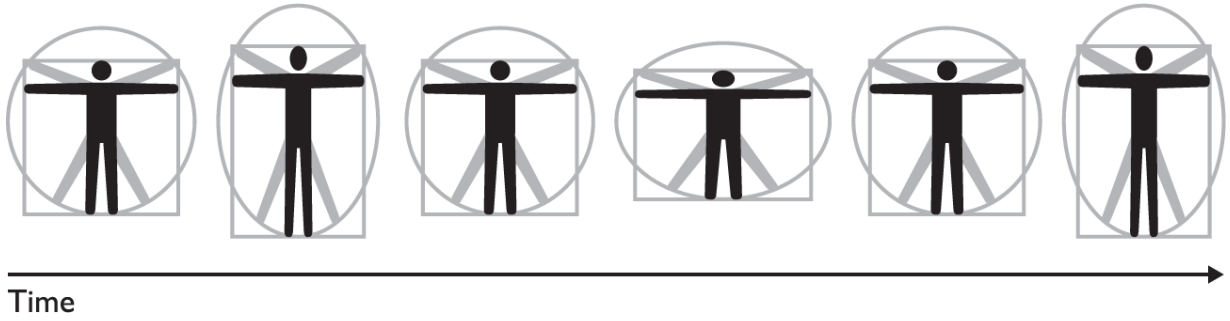


Figure 19: Illustration of the effect of a passing gravitational wave. As a gravitational wave hits face-on, it stretches the space it's moving through vertically while squeezing it horizontally, and then vice versa, with each wave crest. If you are in the path of the wave, you are alternately a bit taller and thinner, and a bit shorter and wider, over and over until the wave passes. The magnitude of the stretching of your body is only about one millionth the width of a proton.

We have good reasons for being suspicious. Compared to the other forces, gravity is an oddball. Not only does it look totally different from a mathematical point of view, it's way too weak. Sure, when you get together enough mass for a galaxy, or a black hole, it seems fairly strong. But in daily life, it's easily the weakest force you encounter. Every time you lift a coffee cup you're overcoming the gravitational pull of the entire planet. It takes putting the mass of the Sun into something the size of a city before gravity can even begin to compete with the atomic and nuclear forces holding atoms together.